

Heat supply and host-sector demand are not yet reconciled

Hydrogen burned in a gas turbine is correctly identified as **heat supply**, even when the IAM demand originates in the chemicals sector. The current model, however, uses one sector label both to estimate physical demand nodes and to choose the H2 distribution market. Those are different decisions.

CURRENT INTERPRETATION	A chemical-sector gas turbine consumes the Heating H2 market because hydrogen is used as an energy carrier for heat, not as chemical feedstock.
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The missing distinction

IAM demand	Physical context	End use	H2 market
Industry - Chemicals - H2	Chemical plant node estimation	Heat supply gas turbine / burner	Heating distribution option

Why this is a limitation

- Demand aggregation currently classifies the IAM coordinate as **Chemicals**.
- Consumer routing classifies the gas-turbine proxy as **Heating**.
- Heating demand and its 100% pipeline rule currently represent buildings, not industrial heat.
- A Chemicals market may therefore be sized from thermal demand while the corresponding burner uses Heating.

Proposed future resolution

- Keep **sector/subsector** for the host site and demand-node method.
- Add an explicit **hydrogen_end_use** or **hydrogen_market** field.
- Assign thermal chemical demand to Heating while retaining chemical-plant node information.
- Generate Chemicals markets only from explicit feedstock or non-thermal process demand.

DECISION STILL REQUIRED	Confirm whether industrial H2 final-energy variables represent heat only, and whether the Heating distribution rule should be universal or vary with the host sector. Until then, strict reconciliation between sector demand, logistics, and consumer links should not be claimed.
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Scope of the current code change: Transport consumer routing is corrected separately. This note records the heating/host-sector inconsistency for a later modelling decision; it does not prescribe or implement that larger change.